

RAM CHARAN VANGA

RA2511027010164

Subject : DSA

Unit 2, S3, SLO2 - Introduction to Linked Lists

1. A linked list, in simple terms, is a _____ of data elements.

Answer: linear collection / sequence

2. The data elements in linked lists are called _____.

Answer: nodes

3. Linked lists acts as a _____ to implement data structures such as stacks, queues, and their variations

Answer: fundamental underlying data structure

4. A linked list can be perceived as a _____ in which each node contains one or more data fields and a pointer to the next node.

Answer: chain of nodes

5. In a linked list, every node contains a pointer to another node which is of the same type, it is also called a _____.

Answer: self-referential structure

6. Linked lists contain a pointer variable START that stores the _____ in the list.

Answer: address of the first node (head)

7. We can traverse the entire list using _____ which contains the address of the first node; the next part of the first node in turn stores the address of its succeeding node

Answer: START (or pointer variable PTR)

8. If START = NULL, then the linked _____.

Answer: list is empty

9. Linked lists provide an efficient way of storing _____ and perform basic operations such as insertion, deletion, and updation of information at the cost of extra space required for storing address of the next node.

Answer: dynamic data

10. In C, we can implement a linked list using the following code:

struct node

```
{  
int ____;  
struct _____;  
};
```

Answer: struct node {

```
int data;  
struct node *next;  
};
```